Define Problem Statements

Problem framing and measurable outcomes

Project: AI WeatherWise API   Track: AI-ML & GEN-AI Track

Prepared: 07 October 2026   Team ID: Not provided

Maximum Marks: 5

Team Members & Assigned Roles

Team Member	Role
Jai Bharath A	Team Lead & Lead Backend/API Architect
Mukesh Balaji	Generative AI & Prompt Engineering Engineer
Yadagiri Mani	Database & Data Engineering Specialist
Deepak G	Security, Testing & Reliability Engineer
Naveen	API Integration, Documentation & Demonstration Engineer

Primary Problem Statement

Existing weather interfaces often expose isolated numbers—temperature, humidity, wind, pressure and precipitation—without translating them into simple, personalized actions. Users must interpret the data themselves.

Problem	Who is affected?	Impact	WeatherWise response
Data overload	Everyday users	Hard to interpret conditions	Natural-language AI summary
Generic advice	Commuters/students	Advice is not contextual	Outfit, commute and hydration guidance
API dependence	Application owners	External outages break features	Deterministic mock/rule fallbacks
Unsafe saved data	Authenticated users	Cross-user access risk	JWT guards + ownership checks + compound index

Success Criteria

- 10 documented REST endpoints covering authentication, weather, locations and AI intelligence.
- AI responses remain understandable and actionable rather than repeating raw metrics.
- Protected location data is isolated by authenticated user.
- Weather and AI failures degrade gracefully through deterministic local fallbacks.
- Testing evidence covers functional correctness, response latency and failure scenarios.